

RG11

## Minerals, Energy, Tourism and Transport of Rajasthan

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*Expect 3-5 questions a paper from this chapter alone. The certainties are: one mineral-to-location matching set, one thermal-station-to-district item, one solar or renewable energy item, and increasingly one on the Pachpadra refinery. Tourism and transport supply another one in most papers, usually the UNESCO list or the Jaipur Metro.*

*This is the biggest single chapter in Rajasthan Geography, and it contains the most repeated question type in the whole subject - match the mineral to the mine to the district. RESEARCH records mineral-mine matching as asked in every single paper. Learn the mine list the way you learn a multiplication table, then add energy, tourism and transport on top.*

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## Rajasthan - Minerals, Energy, Tourism & Transport

### India's **SOLE** producer

- Lead-zinc ore
- Wollastonite
- Selenite
- Jasper

### India's **LARGEST** producer

- Gypsum
- Rock phosphate
- Silver
- Marble
- Calcite, soapstone, ball clay, sandstone

### Metallic - mine to district

- Zawar, Rampura-Agucha, Rajpura-Dariba - lead-zinc
- Khetri-Singhana, Kolihaan - copper (Jhunjhunu)
- Degana (Nagaur) - tungsten
- Bhukia-Jagpura (Banswara) - gold
- Rohil, Sikar - uranium

### Non-metallic

- Jhamar Kotra (Udaipur) - rock phosphate
- Got-Manglodh (Nagaur) - gypsum
- Makrana - white marble
- Sanu (Jaisalmer) - SMS limestone
- Mando ki Pal (Dungarpur) - fluorite
- Khera/Belka Pahar (Sirohi) - wollastonite

### Fuel minerals

- Lignite - Palana, Barsingsar, Kapurdi, Jalipa, Giral
- Oil - Mangala, Bhagyam, Aishwariya (Barmer-Sanchore)
- Gas - Tanot, Dandewala (Jaisalmer)
- Pachpadra refinery, HRRL, Balotra

### Power stations

- Suratgarh - Sri Ganganagar (largest)
- Kota Super Thermal - first coal station
- Chhabra - Baran; Kalisindh
- Jhalawar
- Rawatbhata - Chittorgarh, nuclear

### Renewables

- Bhadla Solar Park - Phalodi, India's largest
- 1st in solar, 2nd in total renewables
- RRECL is the nodal agency
- PM-KUSUM, PM Surya Ghar

### Tourism & transport

- 4 UNESCO sites; 6 Hill Forts
- RTDC, Palace on Wheels 1982
- North Western Railway, HQ Jaipur
- Jaipur Metro 2015; Delhi-Mumbai Expressway

## Rajasthan's mineral profile - why it is called the mineral museum of India

Rajasthan produces more varieties of minerals than any other state. It does not lead in coal or iron, so it is not the biggest by value, but in the sheer spread of minerals it is unmatched. For the exam the important thing is a very short list: the minerals for which Rajasthan is India's ONLY source, and the ones where it is India's largest producer. Every year one question comes straight out of those two lists.

**The two lists that must be memorised exactly**

| Rajasthan is India's SOLE producer of | Rajasthan is India's LARGEST producer of |
|---------------------------------------|------------------------------------------|
| Lead-zinc ore                         | Gypsum                                   |
| Wollastonite                          | Rock phosphate                           |
| Selenite                              | Silver                                   |
| Jasper                                | Marble                                   |
| (only these four)                     | Calcite, soapstone, ball clay, sandstone |

- Do not confuse: bauxite is NOT a Rajasthan mineral - it comes from Odisha, Gujarat and Jharkhand.
- Ordinary coal is not a Rajasthan mineral either; what Rajasthan has is LIGNITE, brown coal.
- Both are planted in 'which is NOT produced in Rajasthan' questions.
- Iron ore exists in Rajasthan but in small, low-grade deposits - the state is not an iron producer of rank.

One structural point that helps you remember locations. Metallic minerals in Rajasthan sit along the Aravalli - Udaipur, Rajsamand, Bhilwara, Chittorgarh, Sirohi, Jhunjhunu, Alwar. Non-metallic and evaporite minerals sit in the desert

and the Nagaur-Jaisalmer belt - gypsum, limestone, selenite, bentonite. Fuel minerals sit in the far west - Bikaner and Barmer. If you forget a specific mine, this rule usually gets you the right district family. And for the sole-producer four, remember the line 'Zawar sells jasper and wool' - Zinc-lead ore, SElenite, JASPER, WOLLastonite.

## Metallic minerals - the mine-to-district table you cannot afford to miss

Read the table, then close the book and write the mine names against the districts. This exact set is what RPSC turns into a four-pair matching question, and RESEARCH shows it has been asked in every recent paper. Do not try to remember the geology. Remember mine - district - mineral, in that order.

### Metallic minerals - mine, district, mineral (part 1)

| Mine / belt                           | District  | Mineral                                              |
|---------------------------------------|-----------|------------------------------------------------------|
| Zawar                                 | Udaipur   | Lead-zinc - world's oldest known zinc-smelting site  |
| Rampura-Agucha                        | Bhilwara  | Lead-zinc - India's LARGEST zinc mine                |
| Rajpura-Dariba                        | Rajsamand | Lead-zinc                                            |
| Sindesar Khurd                        | Rajsamand | Lead-zinc - India's largest SILVER-producing mine    |
| Khetri-Singhana, Madan-Kudan, Kolihan | Jhunjhunu | Copper - Khetri Copper Complex, Hindustan Copper Ltd |
| Kho-Dariba                            | Alwar     | Copper                                               |
| Deri-Basantgarh                       | Sirohi    | Copper                                               |

### Metallic minerals - mine, district, mineral (part 2)

| Mine / belt                   | District | Mineral                                    |
|-------------------------------|----------|--------------------------------------------|
| Morija-Banola                 | Jaipur   | Iron ore                                   |
| Neemla-Rajpura                | Dausa    | Iron ore                                   |
| Nathara-ki-Pal, Dalani        | Udaipur  | Iron ore                                   |
| Kalakhunta, Lilwani, Timamori | Banswara | Manganese                                  |
| Degana (Rewat hill)           | Nagaur   | Tungsten (wolframite)                      |
| Bhukia-Jagpura                | Banswara | Gold - the state's principal gold prospect |
| Rohil, in the Khandela belt   | Sikar    | Uranium                                    |
| Jahaj                         | Bhilwara | Uranium                                    |

### Zawar vs Rampura-Agucha - the two lead-zinc names RPSC swaps

| Zawar                                                                   | Rampura-Agucha                                         |
|-------------------------------------------------------------------------|--------------------------------------------------------|
| Udaipur district                                                        | Bhilwara district                                      |
| Famous for ANTIQUITY - zinc distillation from about the 13th century CE | Famous for SCALE - India's largest zinc-producing mine |
| World's oldest known zinc-smelting site                                 | A modern opencast-to-underground operation             |
| Still worked by Hindustan Zinc Ltd                                      | Also worked by Hindustan Zinc Ltd                      |

#### ASKED IN RAS

Asked in RAS Pre 2016 and again in 2024 - the Khetri copper belt lies in which district? Answer: Jhunjhunu, worked by Hindustan Copper Ltd's Khetri Copper Complex. Asked in RAS Pre 2018 - Degana (Rewat hill) in Nagaur is known for which mineral? Answer: tungsten.

- Zawar is in UDAIPUR, not Chittorgarh - Chittorgarh has the Chanderiya smelter, a different thing.
- Rampura-Agucha is in BHILWARA; Rajsamand has Rajpura-Dariba and Sindesar Khurd.
- Silver is not mined as silver - it comes out of the Sindesar Khurd lead-zinc ore.

- Copper is Jhunjhunu first; Alwar and Sirohi are the secondary copper districts.

## Non-metallic and industrial minerals - the second big matching set

If the metallic table is the first favourite, this is the second, and one pair inside it - Jhamar Kotra with rock phosphate - is probably the single most asked geography fact about Rajasthan. Jhamar Kotra in Udaipur is India's largest rock phosphate deposit. Got-Manglodh in Nagaur is the gypsum name. Those two, plus Khetri and Zawar, form the four-pair matching set RPSC has used repeatedly.

### Non-metallic minerals - mine, district, mineral (part 1)

| Mine / belt               | District                            | Mineral                                                   |
|---------------------------|-------------------------------------|-----------------------------------------------------------|
| Jhamar Kotra              | Udaipur                             | Rock phosphate - INDIA'S LARGEST deposit, worked by RSMML |
| Maton, Kanpur-Dakan Kotra | Udaipur                             | Rock phosphate                                            |
| Birmania                  | Jaisalmer                           | Rock phosphate                                            |
| Got-Manglodh, Bhadwasi    | Nagaur                              | Gypsum - Rajasthan produces the bulk of India's gypsum    |
| Sanu                      | Jaisalmer                           | Limestone - high-grade SMS variety for steel plants       |
| Gotan, Mundwa             | Nagaur                              | Limestone (cement grade)                                  |
| Makrana                   | Nagaur region, now Didwana-Kuchaman | White marble - used in the Taj Mahal                      |
| Rajsamand district        | Rajsamand                           | Marble - the LARGEST marble-producing district            |

### Non-metallic minerals - mine, district, mineral (part 2)

| Mine / belt           | District                   | Mineral                                               |
|-----------------------|----------------------------|-------------------------------------------------------|
| Andhi                 | Jaipur                     | Green marble                                          |
| Babarmal              | Udaipur                    | Pink marble                                           |
| Bansi Paharpur        | Bharatpur region, now Deeg | Red sandstone                                         |
| Dholpur and Karauli   | Dholpur, Karauli           | Red sandstone; Jodhpur gives 'chhitar' sandstone      |
| Jalore                | Jalore                     | Granite - the 'granite city of Rajasthan'             |
| Mando ki Pal          | Dungarpur                  | Fluorite (fluorspar) - India's leading deposit, RSMML |
| Khera and Belka Pahar | Sirohi                     | Wollastonite - produced only in Rajasthan             |
| Rikhabdeo / Kherwara  | Udaipur                    | Asbestos; also Ajmer, Dungarpur, Rajsamand            |
| Kaliguman-Tekhi belt  | Rajsamand-Ajmer            | Emerald                                               |
| Rajmahal              | Tonk                       | Garnet                                                |
| Barmer and Jaisalmer  | Barmer, Jaisalmer          | Bentonite                                             |
| Bikaner               | Bikaner                    | Ball clay; Nagaur gives selenite                      |

**ASKED IN RAS**

Asked in RAS Pre 2021 - Jhamar Kotra is associated with which mineral?

Answer: rock phosphate. Asked in RAS Pre 2016 and 2021 as a full matching set - Jhamar Kotra to rock phosphate, Got-Manglodh to gypsum, Zawar to lead-zinc, Degana to tungsten. RESEARCH names this the most frequently repeated matching set in RAS geography.

- Fluorite is Dungarpur (Mando ki Pal); wollastonite is Sirohi (Khera, Belka Pahar) - never swap them.

- Makrana now falls in Didwana-Kuchaman district after the 2023 reorganisation; older books say Nagaur.
- Marble: Makrana is famous for white, Rajsamand produces the most, Andhi green, Babarmal pink.
- Red sandstone is Dholpur-Karauli-Bansi Paharpur; Jodhpur's variety is called 'chhitar'.

#### YAAD RAKHO

Yaad rakho the four-pair set as a sentence: 'Jhamar ka Phosphate, Got ka Gypsum, Zawar ka Zinc, Degana ka Tungsten.' Four mines, four minerals, one sentence. If you learn nothing else from this chapter, learn that line.

### Who mines what - RSMML, Hindustan Zinc and the salt lakes

RPSC asks the operator as often as the mine. There are only three names to know, and each owns a clear set of minerals. Both the big state and public-sector mining houses are headquartered at Udaipur, which is itself a question.

#### The mining companies and what they work

| Company                                        | Headquarters      | What it mines / smelts                                                                    |
|------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------|
| RSMML - Rajasthan State Mines and Minerals Ltd | Udaipur           | Rock phosphate (Jhamar Kotra), limestone (Sanu), gypsum, lignite, fluorite (Mando ki Pal) |
| Hindustan Zinc Ltd (HZL)                       | Udaipur           | Lead-zinc and silver - Zawar, Rampura-Agucha, Rajpura-Dariba, Sindesar Khurd              |
| Hindustan Copper Ltd - Khetri Copper Complex   | Khetri, Jhunjhunu | Copper - Khetri-Singhana, Madan-Kudan, Kolihan                                            |

- Hindustan Zinc's smelters: Debari (Udaipur), Chanderiya (Chittorgarh) and Dariba (Rajsamand).



- Rajasthan is India's **THIRD** largest salt producer.
- Salt sources: Sambhar (India's largest inland salt lake), Didwana, Pachpadra and Lunkaransar.
- Sambhar is also a Ramsar site - the same name serves the environment paper.
- RSMML is a state government undertaking; HZL and HCL are of central origin, HZL now under Vedanta.
- Debari, Chanderiya and Dariba are **SMELTERS**, not mines - a common wrong option.
- Keep the categories apart: Khetri a mine complex, Makrana a marble centre, Sambhar a salt source.

## Fuel minerals - lignite, the Barmer oil basin and the Pachpadra refinery

Rajasthan has no ordinary coal. What it has is lignite, or brown coal, in the western districts, and that lignite feeds power stations right at the pithead. Then, from 2004, came the thing that changed the state's economy - the Barmer-Sanchore oil basin. Learn the lignite fields by district first; they are pure recall.

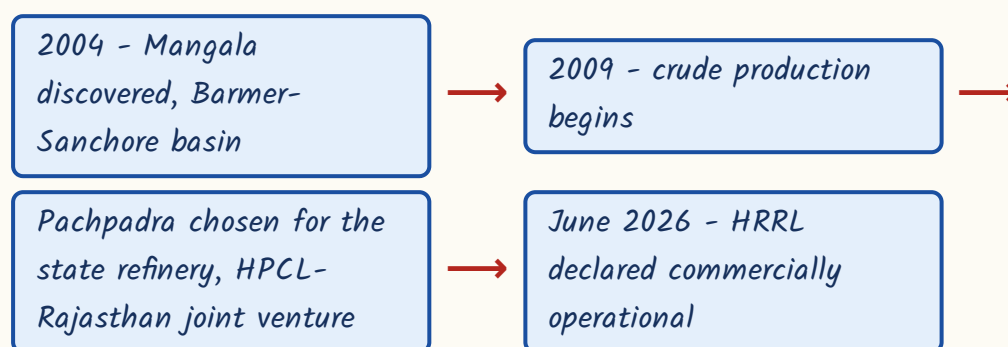
### Lignite fields to districts - a standing matching question

| Lignite field    | District | Note                                           |
|------------------|----------|------------------------------------------------|
| Palana           | Bikaner  | Rajasthan's <b>OLDEST</b> worked lignite field |
| Barsingsar       | Bikaner  | NLC lignite thermal power station here         |
| Kapurdi          | Barmer   | Feeds the Barmer lignite thermal project       |
| Jalipa           | Barmer   | Feeds the Barmer lignite thermal project       |
| Giral            | Barmer   | Giral lignite power project                    |
| Matasukh, Kasnau | Nagaur   | Smaller fields                                 |

Now the oil. In 2004 Cairn Energy struck oil at Mangala in the Barmer-Sanchore basin - the largest onshore discovery in India for decades. The block is RJ-ON-90/1. Production started in 2009, five years after discovery, and that gap between discovery and production is exactly the kind of detail RPSC checks with a wrong statement. The other fields in the same basin are Bhagyam, Aishwariya, Raageshwari and Saraswati. The operator is Cairn, now Vedanta.

- Petroleum - Barmer-Sanchore basin; block RJ-ON-90/1; operator Cairn / Vedanta.
- Mangala discovered 2004; production began 2009. Bhagyam and Aishwariya are the other big fields.
- Natural gas - Jaisalmer basin: Tanot, Dandewala, Bagitibba, Bhagewala, worked by ONGC.
- That Jaisalmer gas feeds the Ramgarh gas-based power station in Jaisalmer district.
- Remember the split: OIL is Barmer, GAS is Jaisalmer.

#### From discovery to refinery - the Barmer story in four steps



### CURRENT LINK

Current link, as of September 2026: the HPCL Rajasthan Refinery Ltd (HRRL) at Pachpadra in BALOTRA district was declared commercially operational in June 2026. It is a 9 MMTPA refinery with a 2.4 MMTPA petrochemical complex. Shareholding is HPCL 74% and the Government of Rajasthan 26%. It is designed to process both Barmer basin crude and imported crude - so it is not an imported-crude-only refinery, which is how a wrong-statement question is built. This is the highest-probability new fact in the whole chapter. Note also that Pachpadra sits in BALOTRA district after the reorganisation, not Barmer district as older books say, and that Pachpadra has two identities - an old salt source and now the refinery site.

## Conventional power - thermal, gas, hydel and nuclear

Thermal station to district is asked as regularly as mineral to district. Four names carry it. Suratgarh is the largest and is nicknamed Rajasthan's thermal power house. Kota Super Thermal was the first coal-based station in the state. Chhabra and Kalisindh are the two Hadoti-region stations that students constantly swap.

### Power stations to districts - learn this cold

| Station                 | District       | Type / note                                                  |
|-------------------------|----------------|--------------------------------------------------------------|
| Suratgarh Super Thermal | Sri Ganganagar | Largest thermal station; 'thermal power house of Rajasthan'  |
| Kota Super Thermal      | Kota           | The state's FIRST coal-based thermal station, on the Chambal |
| Chhabra                 | Baran          | Has both subcritical and supercritical units                 |
| Kalisindh               | Jhalawar       | Two units of 600 MW each                                     |
| Girai lignite           | Barmer         | Lignite-fired                                                |

| Station                                 | District    | Type / note                               |
|-----------------------------------------|-------------|-------------------------------------------|
| Kapurdi-Jalipa (Barmer lignite project) | Barmer      | Lignite-fired                             |
| Barsingsar (NLC)                        | Bikaner     | Lignite-fired                             |
| Ramgarh                                 | Jaisalmer   | Gas-based, fed by Tanot and Dandewala gas |
| Dholpur combined-cycle                  | Dholpur     | Gas-based                                 |
| Rawatbhata Atomic Power Station         | Chittorgarh | Nuclear, on the Chambal                   |

Rawatbhata deserves a line of its own. It stands on the Chambal in Chittorgarh district and it is India's SECOND nuclear power station - Tarapur in Maharashtra was the first. Unit-1 was built with Canadian collaboration. That 'second, not first' point is asked directly. Rajasthan's hydel comes from the same river system: Rana Pratap Sagar and Jawahar Sagar on the Chambal, and Mahi Bajaj Sagar in Banswara.

#### ASKED IN RAS

Asked in RAS Pre 2016 and again in 2023 as a matching set - Suratgarh to Sri Ganganagar, Chhabra to Baran, Kalisindh to Jhalawar, Rawatbhata to Chittorgarh. Asked in RAS Pre 2024 - Kapurdi and Jalipa lignite fields lie in which district? Answer: Barmer.

#### TRAP

Kalisindh is in JHALAWAR, not Kota - the river Kali Sindh flows through Jhalawar. Giral is in Barmer; Barsingsar is in Bikaner - both are lignite, so they get swapped. And Rawatbhata is India's SECOND nuclear station, never the first.

## Renewable energy - where Rajasthan actually leads the country

This is the fastest-moving part of the chapter and RESEARCH says renewable energy in Rajasthan has been asked in every paper since 2016. The logic is simple and worth stating in one line: the Thar gives you the highest solar insolation in India plus vast cheap barren land, so solar goes where the desert is. Jodhpur, Jaisalmer, Bikaner and Barmer carry the state's solar and wind capacity.

- Rajasthan ranks FIRST among Indian states in installed SOLAR capacity - about 41.7 GW as of April 2026.
- But Rajasthan is SECOND in TOTAL renewable capacity, behind Gujarat, as of March 2026.
- Bhadla Solar Park - Bhadla village, PHALODI district (carved out of Jodhpur in 2023).
- Bhadla is about 2,245 MW spread over roughly 56 sq km - INDIA'S LARGEST solar park.
- Bhadla was built in four phases by several agencies - NTPC, SECI, ACME, Fortum, Avaada - not by RRECL alone.
- Wind energy hubs: Jaisalmer, Jodhpur and Barmer; Deogarh is another wind site.

### TRAP

Two live traps. One: Bhadla is INDIA'S largest solar park. It was long described as the world's largest, but it has since been overtaken - as of 2026 say India's largest and you are safe. Two: first in SOLAR is not the same as first in RENEWABLES. Rajasthan is first in solar, second overall behind Gujarat as of March 2026. RPSC will offer both statements and ask which is correct.

### Renewable energy institutions and policies

| Name                                          | What it is                                         | Key detail                                                                                  |
|-----------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------|
| RRECL                                         | Rajasthan Renewable Energy Corporation Ltd, Jaipur | The state NODAL AGENCY for solar, wind and biomass                                          |
| Rajasthan Solar Energy Policy 2019            | State policy                                       | Framework for solar parks and land allotment                                                |
| Rajasthan Integrated Clean Energy Policy 2024 | State policy                                       | The current umbrella clean-energy policy                                                    |
| PM-KUSUM (2019, MNRE)                         | Central scheme for solar in farming                | A: grid plants on barren land; B: standalone solar pumps; C: solarising existing grid pumps |
| PM Surya Ghar Muft Bijli Yojana (Feb 2024)    | Central rooftop solar scheme                       | Up to 300 free units a month; target one crore households nationally                        |

#### ASKED IN RAS

Asked in RAS Pre 2018 and again in 2021 - which is the nodal agency for renewable energy in Rajasthan? Answer: RRECL, the Rajasthan Renewable Energy Corporation Ltd, Jaipur. Asked in RAS Pre 2023 - Rajasthan's rank in installed solar capacity among Indian states? Answer: first.

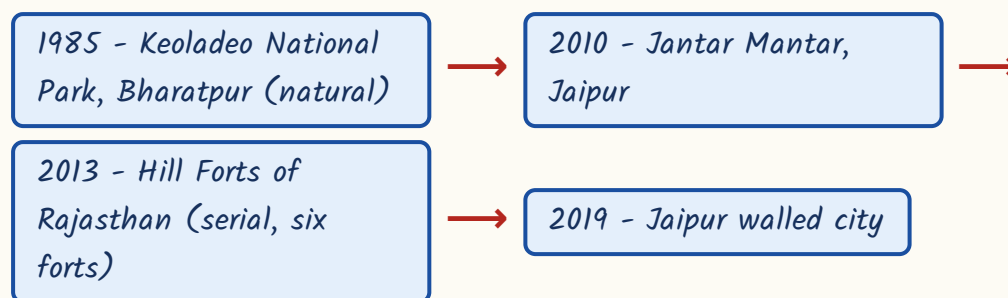
- Power sector structure: RVUNL generates, RVPNL transmits.
- Three discoms distribute - Jaipur, Ajmer and Jodhpur Vidyut Vitran Nigam Ltd.
- Remember the letter: pRasaran = transmission = RVPNL; utpadan = generation = RVUNL.
- PM-KUSUM's three components A, B and C are asked as a match - do not guess the order.

- Hook for PM-KUSUM: A for Area (barren-land plants), B for Bina-grid (standalone pumps), C for Converting existing grid pumps.

## Tourism - the four UNESCO sites and the circuits

Rajasthan has exactly FOUR UNESCO World Heritage inscriptions. Learn them with their years, because RPSC asks the year and asks the count. One is natural, three are cultural. And one of the four - the Hill Forts - is itself a serial property of six forts, which gives a second layer of matching.

### Rajasthan's UNESCO inscriptions in chronological order



### The six Hill Forts of Rajasthan - fort to district

| Fort        | District       | Note                                        |
|-------------|----------------|---------------------------------------------|
| Chittorgarh | Chittorgarh    | The largest fort of the group               |
| Kumbhalgarh | Rajsamand      | Built by Rana Kumbha; the great wall        |
| Ranthambore | Sawai Madhopur | Fort inside the tiger reserve               |
| Gagron      | Jhalawar       | A jal-durg at the Ahu-Kali Sindh confluence |
| Amber       | Jaipur         | Kachhwaha capital before Jaipur city        |
| Jaisalmer   | Jaisalmer      | The living fort of the desert               |

**TRAP**

Mehrangarh (Jodhpur) is NOT one of the six inscribed Hill Forts, however famous it is - that is the standard wrong option. And Rani ki Vav is a stepwell at Patan in GUJARAT, inscribed 2014, not a Rajasthan site. Jaipur's 2019 inscription made it India's second World Heritage city after Ahmedabad.

- RTDC - Rajasthan Tourism Development Corporation runs state tourism; slogan 'Padharo Mhare Desh'.
- Palace on Wheels flagged off 26 January 1982 by RTDC with Indian Railways.
- Royal Rajasthan on Wheels followed in 2009.
- Rajasthan Tourism Policy 2020; tourism was given INDUSTRY status in 2021.
- Heritage hotel scheme converts old havelis and palaces into hotels - a Rajasthan speciality.
- The state's tourist circuits include Desert, Mewar, Vagad, Dhundhar, Godwar, Merwara-Marwar, Brij-Mewat, Hadoti, Shekhawati and NCR.

You do not need every place in every circuit. Learn the anchor of each: Desert = Jaisalmer-Bikaner-Jodhpur-Barmer; Mewar = Udaipur-Chittorgarh-Nathdwara-Kumbhalgarh; Vagad = Dungarpur-Banswara-Beneshwar; Dhundhar = Jaipur-Dausa-Tonk; Godwar = Mount Abu-Ranakpur-Jalore; Hadoti = Kota-Bundi-Jhalawar-Baran; Shekhawati = Sikar-Jhunjhunu-Churu-Mandawa; Brij-Mewat = Alwar-Bharatpur-Deeg-Sariska. The circuit names match the cultural regions you already learnt in the physiography chapter, so this is nearly free.

## Transport - rail, road, metro, air, and the last-minute sweep

Transport gives one clean question most years, and it is almost always a date or a name. Three dates carry it: the North Western Railway created on 1 October



2002, the Jaipur Metro opened on 3 June 2015, and Kishangarh airport starting commercial operations in 2017.

- North Western Railway - headquarters JAIPUR, created 1 October 2002.
- Its four divisions: Jaipur, Jodhpur, Ajmer and Bikaner.
- The Western Dedicated Freight Corridor crosses Rajasthan on its Rewari-Palanpur stretch.
- Jaipur Metro (JMRC) - Phase 1A Mansarovar to Chandpole opened 3 June 2015, the state's FIRST metro.
- Phase 1B extended it underground to Badi Chaupar in 2020; the Pink Line is about 12 km with 11 stations.
- Airports: Jaipur is the main international airport; others at Jodhpur, Udaipur (Dabok), Jaisalmer and Bikaner (Nal).
- Kishangarh in Ajmer district is Rajasthan's GREENFIELD airport, commercial operations from 2017.

### National highways worth knowing by number

| Highway          | Route                                                                                     | Rajasthan stretch                                   |
|------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------|
| NH-48 (old NH-8) | Delhi - Jaipur - Ajmer - Udaipur - Ahmedabad - Mumbai                                     | Part of the Golden Quadrilateral's Delhi-Mumbai leg |
| NH-27            | Porbandar - Silchar (East-West Corridor)                                                  | Enters via Udaipur-Chittorgarh-Kota                 |
| NH-58            | Fatehpur - Nagaur - Ajmer - Beawar - Udaipur - Palanpur; 679 km                           | Runs the length of central Rajasthan                |
| NH-62            | Abohar - Sri Ganganagar - Bikaner - Jodhpur - Pali - Pindwara; 748 km                     | The western north-south spine                       |
| NH-11            | Myajlar (Jaisalmer) - Pokaran - Phalodi - Bikaner - Fatehpur - Jhunjhunu - Rewari; 848 km | Runs across the desert into Haryana                 |

- The Delhi-Mumbai Expressway is about 1,386 km; roughly 373 km of it - the LONGEST state stretch - lies in Rajasthan.
- The Amritsar-Jamnagar Economic Corridor also crosses western Rajasthan.
- Rajasthan's national highway network has crossed 10,000 km; the exact figure moves with each new declaration, so quote the current Economic Review.
- Road DENSITY in Rajasthan is below the national average - the state's huge area is the reason, not a shortage of roads.
- Rajasthan's state highways are managed by the PWD; RIDCOR builds mega-highways on a PPP model.

#### ASKED IN RAS

Asked in RAS Pre 2013 and 2018 - the Mangala oil field belongs to which basin? Answer: the Barmer-Sanchore basin, discovered 2004. Transport and tourism questions in recent papers have run to the UNESCO count (four), the Hill Forts list, and the Jaipur Metro opening date.

#### YAAD RAKHO

Last-minute sweep for this chapter, five lines. Jhamar-Phosphate, Got-Gypsum, Zawar-Zinc, Degana-Tungsten. Suratgarh biggest thermal, Kota first coal, Kalisindh in Jhalawar, Rawatbhata second nuclear. Bhadla in Phalodi, India's largest solar park, RRECL the nodal agency. Four UNESCO sites: 1985, 2010, 2013, 2019. Pachpadra HRRL in Balotra, 9 MMTPA, commercial from June 2026.

#### REVISION RECAP

1. Rajasthan is India's SOLE producer of lead-zinc ore, wollastonite, selenite and jasper.
2. It leads India in gypsum, rock phosphate, silver, marble, calcite, soapstone, ball clay and sandstone.
3. Jhamar Kotra (Udaipur) - India's largest rock phosphate deposit, worked by RSMMML.
4. Got-Manglodh and Bhadwasi (Nagaur) - gypsum. Sanu (Jaisalmer) - SMS-grade limestone.

5. Copper: Khetri-Singhana, Madan-Kudan, Kolihan in Jhunjhunu; also Kho-Dariba (Alwar) and Deri-Basantgarh (Sirohi).
6. Lead-zinc: Zawar (Udaipur, oldest zinc smelting), Rampura-Aguicha (Bhilwara, largest), Rajpura-Dariba and Sindesar Khurd (Rajsamand, largest silver mine).
7. Degana (Nagaur) tungsten; Bhukia-Jagpura (Banswara) gold; Rohil in Sikar and Jahaj in Bhilwara uranium.
8. Makrana white marble (now Didwana-Kuchaman); Rajsamand largest marble producer; Andhi green, Babarmal pink.
9. Mando ki Pal (Dungarpur) fluorite; Khera and Belka Pahar (Sirohi) wollastonite; Jalore is the granite city.
10. RSMML and Hindustan Zinc are both headquartered at Udaipur; HZL smelters at Debari, Chanderiya and Dariba.
11. Lignite: Palana and Barsingsar in Bikaner; Kapurdi, Jalpa and Giral in Barmer; Matasukh in Nagaur. Palana is the oldest.
12. Oil is Barmer-Sanchore - Mangala (2004, production 2009), Bhagyam, Aishwariya, block RJ-ON-90/I. Gas is Jaisalmer - Tanot, Dandewala.
13. HRRL Pachpadra, Balotra district - 9 MMTPA plus 2.4 MMTPA petrochemicals; HPCL 74%, Rajasthan 26%; commercially operational June 2026.
14. Thermal: Suratgarh (Sri Ganganagar) largest, Kota Super Thermal first coal-based, Chhabra (Baran), Kalisindh (Jhalawar).
15. Rawatbhata (Chittorgarh) is India's SECOND nuclear station; hydel at Rana Pratap Sagar, Jawahar Sagar, Mahi Bajaj Sagar.
16. Rajasthan is FIRST in installed solar (about 41.7 GW, April 2026) but SECOND in total renewables behind Gujarat (March 2026).
17. Bhadla Solar Park, Phalodi district - about 2,245 MW over 56 sq km, INDIA'S largest solar park. RRECL is the nodal agency.
18. Policies: Solar Energy Policy 2019, Integrated Clean Energy Policy 2024, PM-KUSUM A/B/C, PM Surya Ghar (300 free units).
19. Four UNESCO sites: Keoladeo 1985, Jantar Mantar 2010, Hill Forts 2013, Jaipur walled city 2019. Six Hill Forts - Chittorgarh, Kumbhalgarh, Ranthambore, Gagron, Amber, Jaisalmer.
20. Transport: North Western Railway HQ Jaipur (1 October 2002), Jaipur Metro 3 June 2015, Kishangarh greenfield airport, 373 km of the Delhi-Mumbai Expressway in Rajasthan.